

A Full Answer to the Weakly Precompact Domination Question

Run fa_banach_001, agent agent_lane_16

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Source Problem

The source paper is Bo Xiang, Jinxi Chen, and Lei Li, *Weak precompactness in Banach lattices*, arXiv:2207.06038. In Remark 3.7, after proving the result for σ -Dedekind complete domain lattices, the authors ask whether the following assertions are equivalent for arbitrary Banach lattices E and F :

- (1) every positive operator $S : E \rightarrow F$ dominated by a weakly precompact positive operator is weakly precompact;
- (2) either
 - (a) the norm of E' is order continuous, or
 - (b) every order interval in F is weakly precompact.

The source paper states that the answer is affirmative when E is σ -Dedekind complete and leaves the general case open.

by a weakly compact positive operator is weakly compact if and only if either E' or F has an order continuous norm. Note that the order continuity of the norm on F is equivalent to the weak compactness of order intervals in F . Naturally, for the domination problem of weakly precompact positive operators the authors pose the following question:

Question. *Are the following assertions are equivalent for a pair of Banach lattices E and F ?*

- (1) *Every positive operator $S : E \rightarrow F$ dominated by a weakly precompact operator is weakly precompact.*
- (2) *One of the following two conditions holds:*
 - (a) *The norm of E' is order continuous.*
 - (b) *Every order interval in F is weakly precompact.*

Clearly, the answer to this question is affirmative when E is σ -Dedekind complete. For the general case we are not able to answer it and have to leave it open.

Proof Intuition

The only missing direction in the source paper is the sufficiency of condition (a) without an extra completeness or weak-order-unit hypothesis on E . Weak precompactness is a sequential property: to show that $S(B_E)$ is weakly precompact, it is enough to start with an arbitrary sequence $(x_n) \subset B_E$ and find a weakly Cauchy subsequence of (Sx_n) .

The sequence (x_n) lives in a much smaller lattice than E : the closed sublattice G generated by the countable set $\{x_n : n \in \mathbb{N}\}$. This G is separable. The hypothesis that E' has order-continuous norm passes to closed sublattices, so G' has order-continuous norm. Separability of G gives a strictly positive functional on G , i.e. a weak order unit in G' . Thus the source paper's Theorem 3.4 applies to the restricted operators $S|_G \leq T|_G$, and it says that $S|_G$ is weakly precompact. Applying this to the original sequence gives the desired weakly Cauchy subsequence.

Auxiliary Lemmas

Lemma 1 (heredity to closed sublattices). *Let E be a Banach lattice whose dual norm is order continuous. If G is a closed sublattice of E , then the norm of G' is order continuous.*

Proof. We use the standard criterion: the norm of X' is order continuous if and only if every norm-bounded disjoint sequence in X is weakly null.

Let (g_n) be a norm-bounded disjoint sequence in G . It is also a norm-bounded disjoint sequence in E , hence it is weakly null in E by the criterion applied to E . If $\gamma \in G'$, extend γ to a functional $\tilde{\gamma} \in E'$ by Hahn–Banach. Then $\gamma(g_n) = \tilde{\gamma}(g_n) \rightarrow 0$. Thus (g_n) is weakly null in G . Applying the same criterion to G shows that the norm of G' is order continuous. $\square$

Lemma 2 (separable lattices have dual weak units). *If G is a separable Banach lattice, then G' has a weak order unit.*

Proof. If $G = \{0\}$ the assertion is trivial. Otherwise choose a norm-dense sequence (u_n) in the positive unit sphere of G . For each n , choose $\varphi_n \in (G')_+$ with $\|\varphi_n\| = 1$ and $\varphi_n(u_n) = 1$. Set

$$\varphi = \sum_{n=1}^{\infty} 2^{-n} \varphi_n \in (G')_+.$$

We claim that φ is strictly positive. Let $0 < g \in G$ and normalize $v = g/\|g\|$. Choose u_n with $\|u_n - v\| < 1/2$. Then

$$\varphi_n(v) \geq \varphi_n(u_n) - \|\varphi_n\| \|u_n - v\| > 1/2,$$

so $\varphi(g) > 0$. A strictly positive functional on G is precisely a weak order unit of the dual Banach lattice G' . $\square$

Main Theorem

Theorem 1. *Let E and F be Banach lattices. The following assertions are equivalent.*

- (1) *For all positive operators $S, T : E \rightarrow F$ with $0 \leq S \leq T$, if T is weakly precompact, then S is weakly precompact.*
- (2) *Either the norm of E' is order continuous or every order interval in F is weakly precompact.*

Proof. The implication (1) $\Rightarrow$ (2) is Theorem 3.6 of Xiang–Chen–Li.

Assume (2). If every order interval in F is weakly precompact, then (1) follows directly from Theorem 3.3 of Xiang–Chen–Li: the solid hull of every weakly precompact set in F is weakly precompact, and $S(B_E) \subseteq \text{Sol}(T(B_E))$.

It remains to handle the case where the norm of E' is order continuous. Let $0 \leq S \leq T : E \rightarrow F$ and suppose that T is weakly precompact. We prove that S is weakly precompact by the sequential definition. Let (x_n) be an arbitrary sequence in B_E , and let G be the closed sublattice of E generated by $\{x_n : n \in \mathbb{N}\}$. Then G is separable. By the first lemma, the norm of G' is order continuous; by the second lemma, G' has a weak order unit.

The restricted operators $S_G = S|_G$ and $T_G = T|_G$ are positive operators from G into F , and $0 \leq S_G \leq T_G$. Moreover T_G is weakly precompact, since

$$T_G(B_G) \subseteq T(B_E).$$

The source paper’s Theorem 3.4 applies to G because G' has order-continuous norm and a weak order unit. Therefore S_G is weakly precompact. Hence the sequence $(Sx_n) = (S_Gx_n)$ has a weakly Cauchy subsequence in F .

Since every sequence in $S(B_E)$ has a weakly Cauchy subsequence, $S(B_E)$ is weakly precompact. Thus S is weakly precompact, proving (1). $\square$

Verification Notes and Limitations

The proof uses no computation. The two added lemmas are standard separable Banach-lattice reductions; the rest of the argument invokes the source paper’s Theorems 3.3, 3.4, and 3.6. The result is a full answer to the question as stated in Remark 3.7.

Bounded novelty check: the run’s registry, solution, attempt, and proof-gap indexes were searched for arXiv:2207.06038 and core phrases including “weakly precompact”, “dominated by a weakly precompact operator”, and “order interval”. Web searches on 2026-06-28 for the exact title and exact domination-problem phrases returned the source arXiv record and no later independent resolution.

Human Review Recommendation

The packet should be reviewed as a likely valid full solution. The main point for review is the heredity lemma for order-continuity of the dual norm; once that standard criterion is accepted, the proof is a direct separable localization of the source paper’s own theorem.

References

- [1] Bo Xiang, Jinxi Chen, and Lei Li, *Weak precompactness in Banach lattices*, arXiv:2207.06038, 2022.
- [2] Charalambos D. Aliprantis and Owen Burkinshaw, *Positive Operators*, Springer, 2006.
- [3] Peter Meyer-Nieberg, *Banach Lattices*, Universitext, Springer, 1991.